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Project Title: A Mobile Application based on Explainable Artificial Intelligence for Intelligent Health Communication on Vaccine Related Narratives

Abstract

Digitalization and artificial intelligence (AI) have transformed health communication and information processing across sectors. This study proposes mobile application based on Explainable Artificial Intelligence (XAI) to enhance the spread of dominant vaccine-related narratives in health communication channels in Rwanda by containing them in a mobile application. The proposed mobile application integrates Natural Language Processing (NLP), machine learning (ML), and explainability methods to process communication data and extract dominant narratives related to vaccination. The use of XAI contributes to the explainability of the content in the app by generating various alternatives of paraphrased texts to enhance the communication. This study will evaluate the contribution of the app by conducting field study on randomly selected target population to compare their perspective on the vaccine-related narratives before and after the use of the proposed mobile application. The evaluations metrics for the population perspective will be the vaccine knowledge, attitudes, trust, and behavioral intention.

Background

In technologically advanced countries such as the United States, Canada, and Australia, AI-driven applications are increasingly getting integrated into digital health communication services to improve their personalization, transparency, and stakeholders' trust (Malik et al., 2026; Schmidt et al., 2025). Various technologies that reshaped health communication started with inventions in various industrial revolutions (Industry 1.0 up to the current Industry 5.0 led by Human-Centric AI Era) (Ashiq, 2026; Kumar et al., 2025). Industry 5.0 inventions made digitalization a key tool to transform all works towards information processing and disseminating in contemporary society (Ashiq, 2026). The rapid evolution of Artificial

Intelligence (AI) has accelerated services digitalization (some being in web applications and others mostly using the mobile applications (mobile apps) enabled by an increased availability of mobile devices) by enabling applications to learn from large-scale data to suggest the possible nuances and guide in interpretation (Bădoi et al., 2026).

Challenges in disseminating and interpreting health-based information such as vaccine-related narratives remain a significant barrier to implementation of various health programs such as vaccine acceptance. Variations in digital content provided and the spreading approach in line with health system implementation, especially for the newly arising programs face a random interpretation sometimes in sub-standardized perspectives in relation to what the programs implementers expect (Schmidt et al., 2025). Although large volumes of health-related data are generated through community communication channels and some digital platforms, the transformation of such data into actionable and understandable insights remains limited. Consequently, health narratives are often interpreted inconsistently, creating gaps in communication, public understanding, and program implementation. These limitations are also found in vaccine uptake program, as it is among the crucial health programs, highlight the need for the state-of-the art technologies such as a mobile application to enhance their dissemination to the communities and other stakeholders on the propgrams.

Unlike conventional digital health messages spread based on the black-box AI models, XAI-based digital applications generate interpretable outputs that reveal the reasoning behind model predictions and recommendations. By combining XAI with Natural Language Processing and machine learning techniques, it becomes possible to analyze the narratives and give insights on the stakeholders' perspectives. In the context of vaccine acceptance, the proposed mobile application is built based on XAI to assist in interpreting health communication narratives and produce insights with varied related words but having a same meaning to enhance the community understanding (Delelegn & Tesfachew, 2021; Dubé et al., 2013; Gellin et al., 2000).

Research gap

In spite of AI systems that have accurate data-driven decision-making, most existing models remain based on black-boxes in nature that limit their interpretability, particularly in real-world health-based communication context. In current implementations, most AI systems prioritize accuracy over transparency, resulting in reduced trust, limited user adoption, and weak

accountability (Karami, 2026; Luong et al., 2026). This gap is critical in health communication systems where decisions must be understandable to both technical and non-technical users.

Despite the rapid advancement of AI technologies, there remains a gap in their adaptation to contribute to intelligent health communication systems that limit the contribution of AI in alleviating the misinterpretation in dynamic health information ecosystems.

Aim and objectives

In this work we aim at providing a mobile application using explainable AI combined with natural language processing and machine learning, to enhance intelligent health communication on vaccine related narratives.

The specific objectives are:

1. To collect vaccine-related contextual data from Rwanda health community communication channels and process the collected data to be integrated in the proposed mobile application.
2. To integrate in the mobile application machine learning and XAI capabilities to interpret the contextual data from objective one and give context-aware insights with for health communication.
3. To evaluate the effectiveness of the use of the mobile application by evaluating the vaccine-related knowledge, attitudes, trust, and behavioral intention among target communities with and without the use of the application.

Research question(s)

Primary Question: How a mobile application based on integrated XAI framework combined with NLP operates to bridge the gaps between the vaccine experts' expectations and the community awareness on vaccines in Rwanda?

Research outputs

To answer the above research question, the following deliverables will be provided:

1. A dataset describing dominant vaccination-related narratives extracted from Rwanda health communication channels and digitized vaccine-related content possible to be integrated in any digital platform.
2. A mobile application integrated machine learning and XAI for interpreting the vaccine-related dataset obtained in deliverable one.

3. An evaluation report on the effect of the mobile application in deliverable two use on vaccine-related knowledge, trust, and behavioral intentions among target communities in Rwanda.

Conceptual framework

This study integrates the Health Belief Model (HBM) with a Human-in-the-Loop Explainable Artificial Intelligence (XAI) framework. The Health Belief Model explains health-related behavior through perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cues to action, and self-efficacy (Haller et al., 2008). In this framework, these constructs are operationalized through computational analysis of health communication data. The XAI system processes textual data to extract behavioral and informational patterns corresponding to HBM constructs (Luong et al., 2026). NLP identifies dominant narratives and sentiments patterns, and machine learning models map these patterns to behavioral (Delelegn & Tesfachew, 2021). Integrated into a mobile application, the explainability component provides interpretable outputs in different meanings of the word to enhance the understanding of the main word. A Human-in-the-Loop mechanism ensures that AI-generated outputs from the mobile application are reviewed and validated by Human (In the context of this work this will be validated by the target group from the community health workers and health experts) and feedback from human reviewers is incorporated into the system to refine subsequent analyses and recommendations. The research process will follow a three-phase workflow aligned with the expected research outputs.

Phase 1: Narrative Dataset Development

This phase focuses on collecting and preparing vaccination-related communication data from Rwanda health communication channels, including Focus Group Discussions (FGDs) and Key Informant Interviews (KIIs) from the community-based health workers and health experts.

Phase 2: Mobile Application Development

The second phase utilizes the dataset generated in Phase 1 to develop mobile application with an integrated ML and XAI framework. ML will be employed to interpret and categorize the extracted narratives, while XAI mechanisms will provide transparent explanations of model outputs. The linguist will be consulted to work on validating the generated interpretations and refining communication recommendations.

Phase 3: Mobile Application Deployment and Evaluation

The final phase will involve deploying the Mobile application and conduct a study on two groups, namely, a group that will be using the existing system to teach the vaccine-related content, and another group that will be using the proposed mobile application. The aim of the study will be the evaluation of the effect of using the proposed mobile application in enhancing health communication in ration to vaccines.

Methodology / Research Design

This study adopts a sequential mixed-methods research design conducted over 12 months and organized into four structured phases. The design integrates quantitative and qualitative approaches to support the development, adaptation, and evaluation of and the proposed mobile application for health communication.

Phase 1: Formative Research (Months 1–3)

1. Collect existing health communication narratives related to the vaccines in Rwanda from official community health channels, including Ministry of Health materials, CHW reports, and community dialogue transcripts.
2. Conduct Focus Group Discussions (FGDs) (5–8 sessions) and Key Informant Interviews (KIIs) (8–10 participants) involving community members, CHWs, and health professionals in relation to the vaccine-related narratives in the state-of-the art.

Phase 2: Proposed Mobile Application Tailoring to the Vaccine-related Narratives (Months 4–6)

1. Working on the NLP model for converting the textual data into the machine understandable formats to be further use to feed and train the mobile application.
2. Tailoring the design of the mobile application to the interfaces and content accommodating the vaccine-related narratives.
3. Conduct co-design a workshop with CHWs and public health experts to validate mobile application's outputs and refine its usability.

Phase 3: Pilot Intervention (Months 7–9)

1. Implement a quasi-experimental design in selected community groups.
2. Train a group of community health workers on the use of the mobile application.
3. Collect data from the group that will be using the state-of-the art method and the one that will be using the proposed mobile application from participants.

Phase 4: Evaluation and Dissemination (Months 10–12)

1. Analyze quantitative and qualitative data to assess changes in knowledge, attitudes, trust, and behavioral intention regarding the vaccine acceptance.
2. Evaluate the proposed mobile application performance, interpretability, and CHW usability feedback.
3. Disseminate findings through a peer-reviewed journal publication.

Ethical Considerations

So far, the envisaged research doesn't present any ethical limitation but in case this gets needed, a full ethical approval will be obtained from the University of Rwanda and relevant institutions. Informed consent will be secured for FGDs and KIIs. Participants will be assured of confidentiality and the right to withdraw.

Scope

The study will be conducted in Rwanda as geographic area targeting routine childhood immunizations and potential pandemic vaccines. CHWs are the primary users of mobile applications, ensuring community-level relevance. The project's scope is limited to a pilot evaluation and does not include a full-scale national rollout, which would be the subject of future research.

Limitations and Mitigation Strategies

Limitations:

The study acknowledges several potential limitations. Firstly, the success of the intervention relies on CHW adoption and digital literacy. Secondly, the short duration of the pilot may not be sufficient to observe long-term behavioral changes in vaccine uptake.

Mitigation Strategies

1. *CHW digital literacy*: Mitigated by targeted training tailored to the use of the proposed mobile application and simplified interfaces.
2. *Short pilot duration*: Focused on immediate outcomes (knowledge, attitudes, intentions).

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